SemanticSplat: Graph-Pruned Semantic Search

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Abstract

Digital twins based on 3D Gaussian Splatting (3DGS) are photorealistic but do not directly provide the semantic structure needed for natural-language queries. We present SemanticSplat, a representation-agnostic query layer that organizes captured evidence into a scene–zone–region–entity–view hierarchy and uses top-down pruning before object or view grounding. The central experiment controls the semantic map: graph and flat variants use identical records, queries, labels, and scorers, differing only in candidate selection. Across five manually captured indoor and outdoor scenes (97 RGB-D views, 1,066 semantic items, 150 queries), graph search reduces mean views checked from 19.4 to 4.77 (75.5%) and estimated input context from 3,168 to 1,014 tokens (68.2%). On 125 queries with verified view labels, graph hit@1 is 0.680 vs. 0.768 for flat search and hit@3 is 0.808 vs. 0.928, exposing a quality–cost trade-off under the current deterministic lexical ranker. BBQ-aligned oracle-map pilots add 56 Replica and 48 ScanNet queries: graph search matches flat lexical quality while checking 83.2% and 77.1% fewer objects, respectively. We also report native predicted-map ConceptGraphs runs on all eight selected scenes of each dataset and a full hardware-adapted LangSplat pipeline on one ScanNet scene. These external executions use different map-construction protocols and are diagnostic references, not a cross-method ranking. Code, schemas, frozen outputs, and figure sources accompany the preprint.

1 Introduction

3D Gaussian Splatting [12] provides real-time, photorealistic novel-view rendering, but its geometric and appearance parameters do not by themselves say that a region is a lobby, that an object is a chair, or that a view supports the request "find a place to sit near reception." Digital-twin interfaces, robots, and AR systems therefore need a semantic layer that can retrieve grounded evidence without serializing every observation for every query.

Flat search is a strong and transparent baseline: rank every semantic record and return the best match. Its query cost, however, grows with the complete map, and unrelated rooms repeatedly enter the context. Language-field methods instead distill vision-language features into radiance fields or Gaussians [13, 17]; object-centric methods construct open-vocabulary 3D graphs [7, 15]. SemanticSplatstudies a complementary question: how much context can a semantic hierarchy remove when graph and flat search are forced to use exactly the same records and scorer, and what quality is lost by pruning?

Contributions.

- A queryable semantic-map contract for 3DGS-backed scenes, comprising a canonical entity schema, scene hierarchy, structured query, explicit traversal trace, and grounded result schema.
- A controlled graph-vs.-flat benchmark on five captured scenes and two BBQ-aligned public-dataset pilots, with retrieval, 3D grounding, latency, checked-candidate, context-size, and token metrics kept protocol-specific.
- A reproducible account of one-time graph construction, 10,000-resample paired uncertainty intervals, query-type and scene breakdowns, fallback and relation ablations, and native ConceptGraphs/LangSplat execution evidence.
- Explicit claim boundaries: internal annotations are manual reference labels, public SemanticSplat pilots use oracle semantic candidates, and native external runs are not same-protocol comparisons.

2 Related Work

Language fields and semantic Gaussians. LERF [13] embeds language into a radiance field. LangSplat, Semantic Gaussians, and LEGS attach open-vocabulary features to Gaussian representations [17, 8, 20]. These methods provide dense or point-wise relevance, whereas our measured variable is explicit hierarchical candidate pruning over frozen semantic records.

Open-vocabulary maps and scene graphs. ConceptFusion and OpenScene build open-set or open-vocabulary 3D representations [10, 16]. ConceptGraphs constructs object-centric 3D scene graphs from RGB-D observations [7], while HOV-SG and Hydra add hierarchical or spatial organization for navigation and robotics [19, 9].

BBQ is our closest algorithmic reference: it combines object captions, 3D extents, spatial relations, and deductive reasoning for open-vocabulary grounding [15]. We align public scene subsets and Acc@k reporting with BBQ where feasible, but do not claim a measured BBQ comparison because this release contains no official BBQ execution artifact.

Grounding datasets and evaluation. ScanNet provides annotated RGB-D reconstructions [4]; Replica provides high-quality indoor digital replicas [18]. Nr3D and Sr3D from ReferIt3D support fine-grained object identification in ScanNet scenes [1], and ScanRefer establishes 3D referring-expression localization [2]. OpenLex3D advocates tiered evaluation of open-vocabulary 3D representations [11]. Accordingly, we separate view-label retrieval, oracle-map object grounding, and predicted-map baseline execution instead of merging their metrics.

Context efficiency. Prompt-compression and retrieval-pruning work studies the removal of irrelevant textual evidence [14, 5, 3, 21, 6]. Tokens are not a substitute for 3D grounding quality; we report context and checked candidates alongside hit@k, Recall@1, and 3D IoU-based accuracy.

3 Method

3.1 Hierarchical Semantic Representation

At query time, the system receives a pre-built rooted hierarchy $H = (V, E)$ and canonical semantic records. Each node $v \in V$ stores a stable identifier, node type $\tau(v)$, label, searchable text, attributes, parent/child links, and supporting view IDs. An ordered ancestor path $A(v) = (v_0, \ldots, v_d)$ connects the scene root to a zone, region, entity, and optional view-observation leaf. Containment/evidence edges form the hierarchy; spatial relations remain typed attributes so that they can be validated independently.

The canonical entity candidate is

$$c_i = (\mathrm{id}_i, \tau_i, A_i, y_i, x_i, R_i, B_i, V_i), \tag{1}$$

where y_i is the label, x_i searchable text, R_i typed relations, $B_i = (B_i^{2D}, B_i^{3D})$ optional boxes, and V_i supporting views. Source instance IDs are retained when available; otherwise IDs are derived deterministically. The same c_i records feed graph and flat search.

3.2 Structured Query and Candidate Gating

A natural-language query is parsed as

$$q = (q_t, q_a, r, \mathcal{A}, \iota), \tag{2}$$

with target phrase q_t, optional anchor q_a, relation r, attributes $\mathcal{A}$, and intent/affordance ι. The parser handles simple object, attribute, relational, functional, multi-hop, and negative forms. Deterministic intent expansion maps goals to candidate classes; for example, sit activates chair, sofa, bench, couch, armchair, lounge, and seating.

Table 1: Canonical semantic-input fields. Geometry is optional and validated before spatial scoring.

Field	Meaning
id, type	Stable entity/node identity and semantic type
label, text	Target name and searchable description
ancestors	Ordered scene/zone/region path A_i
attributes	Color, material, affordance, and room cues
relations	Typed target–anchor relation records R_i
bbox_2d/3d	Optional normalized 2D or metric 3D extent B_i
view_ids	Observations supporting the entity V_i

For the internal view benchmark, query tokens Q and branch tokens T_b give the branch score $u_b = |Q \cap T_b|$. If $u_m = \max_b u_b > 0$, graph search retains branches satisfying

$$u_b \geq \max(1, 0.5 u_m). \tag{3}$$

When every overlap is zero, all branches remain eligible. Unzoned entries are never silently dropped. Flat search skips this gate and ranks every searchable record.

For object grounding, Q_e augments the base tokens with aliases and affordance expansions. A branch receives $h_b = |Q_e \cap T_b| / \max(1, |Q_e|)$, and graph search keeps $h_b \geq 0.5 \max_j h_j$. Candidate i is scored as

$$s_i = \min(1, 0.65 e_i + 0.55 \ell_i + 0.20 t_i + 0.25 a_i), \tag{4}$$

where e_i is exact-label match and ℓ_i, t_i, a_i are normalized label, record-text, and synonym/affordance overlaps. Flat lexical applies the identical scorer to the full candidate universe. Stable ID tie-breaking makes every run deterministic.

3.3 Target–Anchor Spatial Reasoning

The parser recognizes near, left/right, above/below, and front/behind in infix and prefix forms. Anchors are retrieved from the complete candidate set, even if the target branch was pruned. A relation can be supported by a stored relation string or geometry. Valid 3D boxes are preferred; normalized 2D boxes are used otherwise. Non-finite coordinates, non-positive extents, or values outside declared bounds invalidate geometric evidence rather than producing a score.

For near, center distance normalized by mean combined box extent must be at most 1.5. Directional predicates compare the relevant center axis with a 5% combined-extent tolerance; front/behind is unresolved for 2D-only geometry. Let $r_i = 1$ for a satisfied relation, -0.35 for a resolved violation, and 0 when unresolved. We update

$$s_i \leftarrow \min(1, \max(0, s_i + 0.35 r_i)). \tag{5}$$

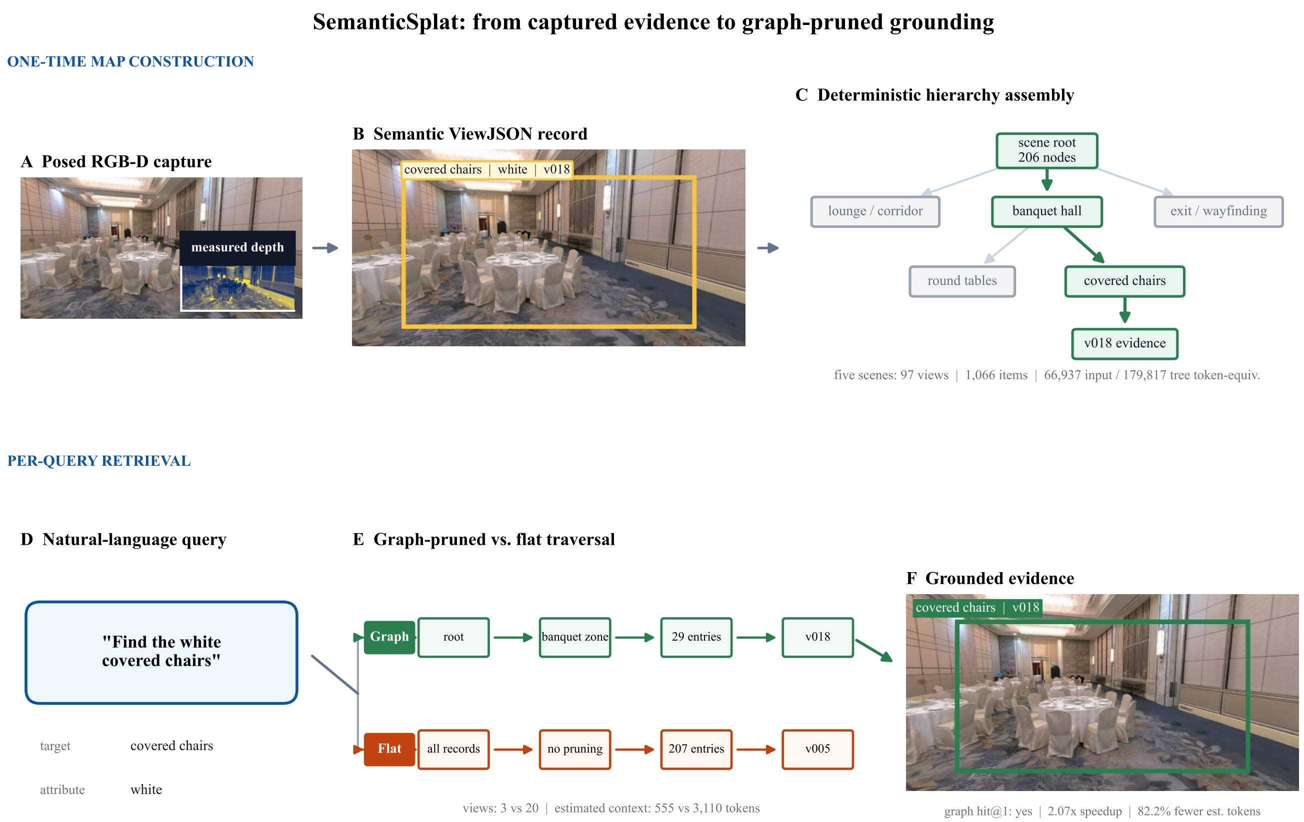

Figure 1: SemanticSplatend to end, assembled from checked-in evidence. Top: captured Conference Hall RGB-D view v018, its ViewJSON annotation, and a hierarchy slice. Bottom: frozen query qv2_010 follows graph-pruned and flat lanes that share the same records and scorer, then returns grounded view evidence. One-time index construction is separated from per-query retrieval. Token values are local serialized-context estimates, not API billing.

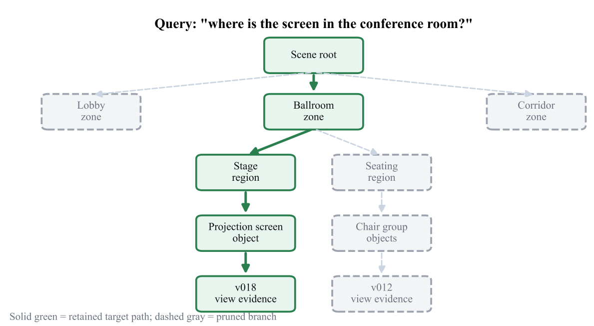

Figure 2: The scene–zone–region–entity–view hierarchy. The solid path is retained for a chair query; gray siblings are pruned before entity scoring.

The trace stores anchor ID, relation score, evidence source, margin, and any box validation failure.

3.4 Fallback, Variants, and Complexity

Graph+fallback expands to every candidate when the best score is below 0.45, the relation is unresolved, or required geometry is rejected. Expansion is recorded explicitly. We evaluate graph, graph+fallback, flat lexical, and flat embedding, with no-hierarchy, no-relation, no-fallback, and flat-only ablations. Flat embedding uses a local BGE-small-en-v1.5 encoder; an unavailable encoder is reported as unavailable rather than replaced silently.

For N searchable records and N_q records retained by the branch gate, flat leaf scoring is $O(N)$ and graph leaf scoring is $O(N_q)$ after a gate over top-level branches. Fallback has the flat worst case. Construction and canonicalization are one-time map costs and are excluded from per-query latency.

3.5 Output and Trace Contract

Every query result contains the selected IDs and box, method/mode, score components, ancestor traversal, checked nodes/objects/views, fallback reason, runtime, serialized context, token measurement metadata, and GT eligibility. The evaluator never converts an unavailable quality field to zero. Table 2 summarizes the contract; the complete JSON schemas and examples are released with the code.

Table 2: Query-result schema groups used for reproducible evaluation.

Group	Required contents
Identity	run/query/scene IDs, method, mode, status
Selection	object/node/view IDs, score, optional 3D box
Trace	visited ancestors, checked counts, fallback reason
Cost	runtime, context characters, token value and tokenizer/type
Quality	GT availability, exact-ID hit, hit@k, IoU and Acc@τ
Provenance	config, map source, model/checkpoint, schema version

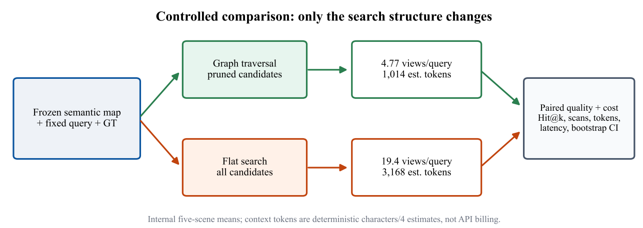

Figure 3: Evaluation tracks. A fair graph-vs.-flat comparison shares maps, queries, labels, and scorer; native external executions have different inputs and do not enter the same ranking.

4 Datasets and Evaluation

4.1 Evaluation Tracks

Figure 3 shows the protocol boundary. Internal graph-vs.-flat results compare candidate selection on manual captured records. Replica and ScanNet SemanticSplat pilots compare variants on official oracle semantic candidates. ConceptGraphs and LangSplat consume their native predicted or learned representations and are therefore reported separately.

Internal digital-twin track. Five team-captured scenes contain 97 RGB-D views, 1,066 semantic items, and 1,064 constructed nodes (Table 3). The benchmark has 150 queries, 25 in each of six query types. Verified expected view IDs exist for 125 queries; the 25 without applicable positive view labels remain in cost metrics but not hit@k denominators. The semantic index and expected labels are manual reference artifacts, not independent public GT.

Public oracle-map tracks. We use the BBQ-aligned Replica subset room0–room2 and office0–office4 (8 scenes, 575 official object boxes, 56 queries), and ScanNet scenes 0011_00, 0030_00, 0046_00, 0086_00, 0222_00, 0378_00, 0389_00, and 0435_00 (392 official object boxes, 48 Nr3D/Sr3D+ queries). Official object identities and 3D boxes form oracle semantic candidates.

Table 3: Internal captured-scene composition.

Scene	Views	Items	Nodes
Conference Hall	20	207	206
Museum	20	225	228
Theater	20	226	226
Outdoor street	21	239	240
Outdoor drone	16	169	164
Total	97	1,066	1,064

These tracks measure retrieval over a known map, not semantic-perception accuracy.

Native external tracks. ConceptGraphs builds predicted object maps for all eight selected ScanNet and eight Replica scenes, then answers 48 and 56 queries. LangSplat completes all native stages and six queries on ScanNet 0011_00 under a hardware-adapted profile. BBQ supplies metric guidance only; no official execution artifact is included.

4.2 Metrics

For method m and query q, we record checked views V_{mq}, objects O_{mq}, nodes G_{mq}, runtime t_{mq}, serialized context characters C_{mq}, and tokens T_{mq}. Internal and oracle-map token estimates are deterministic text-size proxies with their exact rounding rule saved per run; external token values are local tokenizer counts. None are API billing tokens. Relative cost reduction against flat is

$$\Delta_X = 100 \left(1 - \frac{\sum_q X_{gq}}{\sum_q X_{fq}}\right) \%. \tag{6}$$

For verified labels, hit@k is the fraction whose expected view appears in the top k. Public-pilot Recall@1 requires exact target object-ID match. With predicted and reference boxes B_p, B_g,

$$\mathrm{IoU}_{3D} = \frac{|B_p \cap B_g|}{|B_p \cup B_g|}, \qquad \mathrm{Acc}@\tau = \frac{1}{|Q|} \sum_q \mathbf{1}[\mathrm{IoU}_{3D} \geq \tau]. \tag{7}$$

Missing/invalid predictions stay in the eligible denominator with IoU zero. Segmentation mAcc/mIoU/fw-mIoU are not reported without paired predicted segmentation arrays.

Table 4 states the availability rule and denominator for every metric family. Missing reference geometry remains unavailable rather than being converted to a successful zero-distance case. Eligible missed predictions count as zero only where the metric definition requires it.

4.3 One-Time Construction Accounting

The deterministic builder converts 97 manual ViewJSON records into 1,064 tree nodes. Source annotations contain

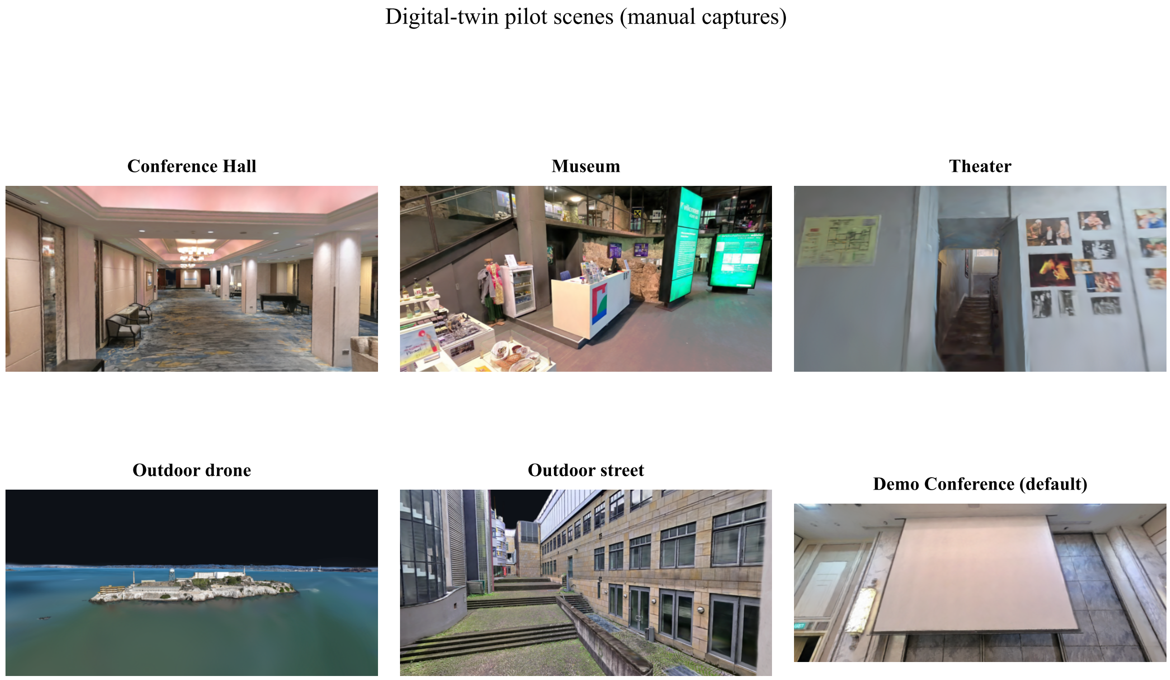

Figure 4: Team-captured indoor and outdoor digital-twin scenes used by the internal track. Licensed public-dataset imagery is not redistributed.

267,877 canonical characters (66,937 character/4 token-equivalents); constructed node records contain 720,910 characters (179,817 token-equivalents). Total serialized input plus output is 988,787 characters, or 246,754 token-equivalents. Across seven clean temporary rebuilds per scene, the sum of per-scene median runtimes is 1.822 seconds. This measures artifact size and local build time; historical human annotation time was not logged and is not inferred.

5 Experiments

5.1 Internal Graph vs. Flat

Across 150 queries, graph traversal checks 4.77 views on average vs. 19.4 for flat (75.5% fewer) and uses 1,014 estimated context tokens vs. 3,168 (68.2% fewer). A paired 10,000-resample query bootstrap gives 95% percentile intervals of [73.2, 77.7]% and [65.3, 71.0]%, respectively. Cumulative context is 152k vs. 475k tokens.

On the 125 queries with verified view labels, graph hit@1 is 0.680 vs. 0.768 for flat and hit@3 is 0.808 vs. 0.928. Paired graph-minus-flat intervals are [-0.184, 0.008] for hit@1 and [-0.184, -0.056] for hit@3. Thus the present lexical hierarchy provides substantial cost control but does not preserve flat hit@k.

5.2 Query-Type and Scene Analysis

Simple-object and functional queries save 74.1% and 71.2% of estimated context, whereas multi-hop and relational queries save 64.3% and 62.6%. Queries spanning multiple entities naturally retain more branches. Savings are positive in every scene: Conference Hall has the largest token reduction (73.2%), while the larger, noisier outdoor-street tree has the smallest (59.1%).

5.3 Public-Dataset Oracle-Map Pilots

Table 7 summarizes the public pilots. On Replica, graph and flat lexical both reach Recall@1/Acc@τ = 1.0 because official identities and boxes are candidates; this is a protocol sanity ceiling, not perception accuracy. Graph checks 12.1 objects vs. 71.9 for flat (83.2% fewer) and uses 303 vs. 1,765 estimated tokens.

On ScanNet, graph matches flat lexical Recall@1 and Acc@0.25 at 0.271 while checking 11.2 vs. 49.0 objects (77.1% fewer) and using 501 vs. 2,117 estimated tokens. Graph+fallback approaches flat cost without improving this lexical quality. Removing relation reranking increases Acc@0.25 from 0.271 to 0.292, identifying the current spatial reranker as a concrete improvement target.

5.4 Native ConceptGraphs and LangSplat Executions

ConceptGraphs builds predicted object maps from 39 selected ScanNet RGB-D views and 40 evenly sampled Replica views. Acc@0.25 is 0.0625 on both tracks; object-label hit / mean 3D IoU is 8/48 / 0.0696 on ScanNet and 7/48 positive queries / 0.0624 on Replica. Returning an

Table 4: Metric prerequisites and denominators. A metric is reported only when the prediction and reference evidence shown below are available.

Metric	Prediction requirement	Reference requirement	Denominator
Efficiency and cost			
Checked views, objects, nodes	Traversal trace	None	All completed queries
Runtime, context, tokens	Timed result and count metadata	None	Results with that measurement
Retrieval quality			
hit@k	Ranked view IDs	Verified expected view IDs	Eligible positive-view queries
Recall@1	Selected object ID	Official expected object ID	Object-ID-eligible queries
Geometric and segmentation quality			
3D IoU	Valid predicted 3D box	Valid official 3D box	Box-eligible predictions
Acc@τ	3D IoU for a prediction	IoU threshold τ	All box-eligible queries; misses score zero
Segmentation mIoU	Predicted per-point labels	Paired GT segmentation	Paired points and classes

Table 5: Measured one-time construction workload for the five captured scenes.

Quantity	Measured value
Manual ViewJSON records	97
Semantic items	1,066
Annotation characters / token-equiv.	267,877 / 66,937
Tree characters / token-equiv.	720,910 / 179,817
Serialized I/O token-equiv.	246,754
Sum of scene median rebuilds	1.822 s

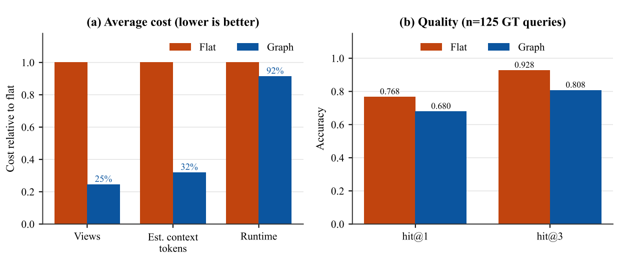

Figure 5: Internal cost and quality. Efficiency uses all 150 queries; hit@k uses the 125 queries with verified view labels.

object for every positive query is retrieval availability, not accuracy.

LangSplat completes RGB 3DGS optimization, SAM/-CLIP preprocessing, autoencoder compression, language-field optimization, rendering, and retrieval on ScanNet 0011_00. It selects an expected view for 4/6 queries, but produces relevance points rather than predicted 3D boxes, so 3D Acc@τ is not applicable. This is a complete native pipeline execution rather than a smoke test. Its hardware-adapted SAM-B/CLIP-B profile uses 3,000 RGB and language iterations on a 6 GB GPU; it does not reproduce the original paper's 30,000-iteration/24 GB scale.

Table 6: Same-input internal benchmark. Runtime is local infrastructure time, not model latency.

Metric	Flat	Graph
Efficiency and cost		
Views/query	19.4	4.77
Est. context tokens/query	3,168.2	1,014.3
Infrastructure elapsed (ms)	6.61	6.05
Retrieval quality ($n = 125$)		
hit@1	0.768	0.680
hit@3	0.928	0.808

Local text-encoder token totals are 627 for ConceptGraphs–ScanNet, 459 for ConceptGraphs–Replica, and 74 for LangSplat. These totals are tied to each native tokenizer and are not directly comparable with character/4 context estimates or API usage.

5.5 Qualitative Cases and Failure Modes

Conference Hall cases show three regimes: isolated screen targets are found in two graph views vs. a 19-view flat scan; a zone-wide projector requires 16 views; and both methods miss an exit-sign query, although graph search stops after three views. Figure 9 grounds the screen query in stage view v018 and exposes a flat full-scan distractor. The cases support cost-control claims, not increased semantic accuracy.

6 Discussion

The same-map experiment isolates hierarchy as the cause of the context reduction: data, labels, and ranking fea-

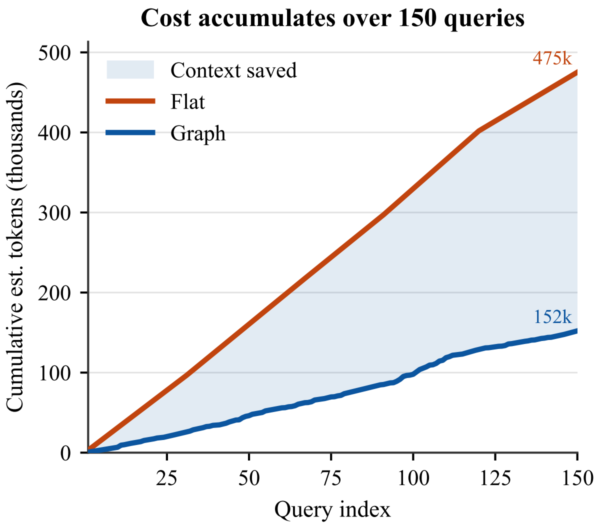

Figure 6: Cumulative estimated context over the 150-query internal benchmark.

Table 7: Oracle-map public pilots. Efficiency is the average number of checked objects. Acc values on these tracks measure retrieval over official object boxes, not semantic map prediction.

Variant	R@1	A@.25	Obj.	Tok.	ms
Replica ($n = 56$)					
Graph	1.000	1.000	12.1	303	0.903
Flat lexical	1.000	1.000	71.9	1,765	8.785
Flat embedding	0.833	0.875	71.9	1,765	137.096
ScanNet ($n = 48$)					
Graph	0.271	0.271	11.2	501	0.785
Graph+fallback	0.271	0.271	47.0	2,027	1.636
Flat lexical	0.271	0.271	49.0	2,117	1.526
Flat embedding	0.250	0.250	49.0	2,117	168.815

tures are held fixed. The internal quality gap shows that hard lexical branch gates can discard a relevant but weakly summarized path. Conversely, public oracle-map pilots show that when candidate labels align with query language, hierarchy can preserve lexical quality while removing most objects. The no-relation ScanNet ablation further shows that relation logic should be confidence-calibrated: unresolved or noisy geometric evidence should not overrule a stronger label match.

The resulting design target is not maximum pruning at any cost. It is a calibrated policy that maximizes candidate reduction subject to a retrieval quality constraint. Soft multi-branch retention, learned summaries, and a budgeted fallback can be evaluated under the same frozen protocol without changing the semantic map.

Table 8: Native external executions. Inputs and protocols differ from the oracle-map SemanticSplat pilots; this table is not a method leaderboard.

Execution	Scenes	Queries	Quality	s/query
CG–ScanNet	8	48	Acc@.25=.063	.336
CG–Replica	8	56	Acc@.25=.063	.274
LangSplat–ScanNet	1	6	view hit=.667	2.037

7 Limitations

- Internal ViewJSON and expected-view labels are manual reference artifacts rather than independent annotations; absolute hit@k can contain correlated labeling bias.
- The internal ranker is deterministic lexical stub logic, not a live provider-backed VLM. Results establish algorithmic and artifact reproducibility, not live-model behavior.
- Replica and ScanNet SemanticSplat pilots use oracle semantic candidates. Their Acc@k values do not measure automatic segmentation or semantic-map construction.
- Relation reranking underperforms its no-relation ablation on the current ScanNet pilot; geometry and relation confidence require calibration.
- Native external runs differ in map construction, frame sampling, output geometry, and resources. LangSplat completes all native stages, but its evidence is one scene under a hardware-adapted 3,000-iteration/6 GB profile; no fair external superiority claim follows from Table 8.
- No official BBQ execution artifact exists; alignment is limited to scenes, query sources, and metrics, so no measured comparison is claimed.
- Historical human annotation time is unavailable. Artifact size and deterministic rebuild time are measured, but person-hours are not inferred.

8 Reproducibility and Release

The release stores aggregate and per-query results, exact scene/query lists, schema versions, run configurations, model/checkpoint provenance, and one-time construction accounting. Figure scripts consume the same frozen JSON used by the consistency checks. Paired confidence intervals use 10,000 query resamples with seed 20,260,715. Licensed ScanNet and Replica scene files are not redistributed; manifests and official access instructions are provided.

Canonical internal results reside under outputs/graph_vs_flat/five_scene_graph_vs_flat_v2; public

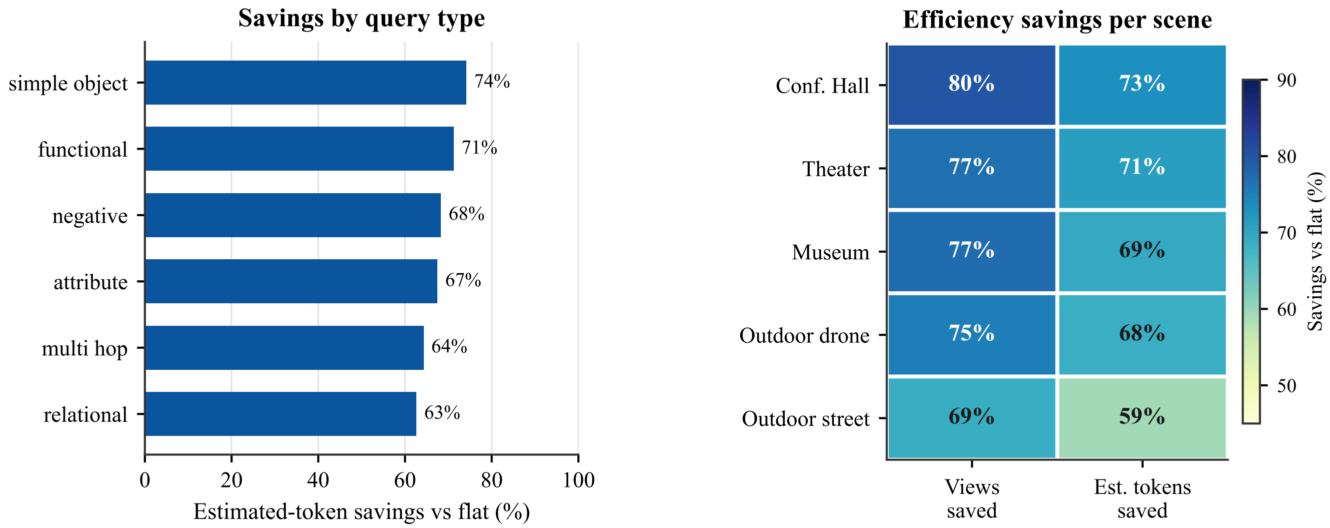

Figure 7: Left: estimated context savings by query type. Right: per-scene view and token reduction. Hierarchy helps most when a query localizes to one branch.

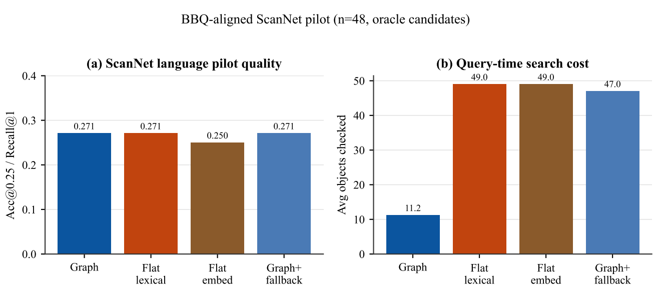

Figure 8: ScanNet oracle-map pilot ($n = 48$). Graph preserves flat-lexical Acc@0.25 while reducing objects and context; fallback approaches flat cost.

pilots under outputs/public_datasets; native runs under outputs/baselines; construction and uncertainty audits under docs/reports/final; and schemas under docs/schemas.

The public artifact includes the backend query implementation, benchmark and adapter scripts, canonical outputs, JSON schemas, this LaTeX source, and an interactive project page. The core verification commands are:

```
python -B -m pytest -q tests -p no:cacheprovider
python -B scripts/check_academic_paper.py
python -B scripts/build_project_site.py
```

9 Conclusion

SemanticSplatdemonstrates that a semantic hierarchy can substantially reduce query-time scene context when evaluated against a flat method on the same map. The five-scene benchmark yields 75.5% fewer checked views and 68.2% fewer estimated context tokens, with a measured hit@k penalty under the current lexical gate. Replica and ScanNet oracle-map pilots preserve flat lexical quality while checking far fewer objects. Native ConceptGraphs runs and the complete LangSplat pipeline execution establish integration coverage but not a fair ranking. Next is calibrated pruning and predicted-map evaluation under a shared scene, query, geometry, and resource protocol.

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Figure 9: Left: room-scoped graph view counts against a 19-view flat scan. Right: grounded stage-screen evidence and a full-scan distractor view.

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